

**YEAR 10 SCIENCE
NATURAL AND PROCESSED MATERIALS
CHEMICAL BONDING - GLOSSARY**

Atom - fundamental building block of matter.

Proton - positively charged particle found in the nucleus of an atom.

Neutron - neutral particle found in the nucleus of an atom.

Electron - negatively charged particle found around the nucleus; is electrostatically attracted to the protons in the nucleus.

Element - substance made up of atoms that are all the same (i.e. same atomic number).

Atomic Number - number of protons in the nucleus.

Atomic Mass - the number of protons and neutrons in the nucleus.

Molecule - two or more atoms chemically combined.

Compound - two or more elements chemically combined in fixed proportions.

Periodic Table - arranges elements in order of increasing atomic number but groups them according to their chemical properties.

Period - a row across the periodic table.

Group - a column going down the periodic table; all elements in a group have the same chemical properties.

Metals - found on the left hand side of the periodic table; donate electrons during reactions.

Malleable - able to be beaten into thin sheets.

Ductile - able to be made (drawn) into a wire.

Lustre - the shine of a metal.

Alloy - a mixture of two or more metals.

Metallic Bond - electrostatic attraction between positive metal ions and delocalised electrons.

Non-metals - found on the right hand side of the periodic table; accept electrons during reactions.

Alkali Metals - belong to Group I of the periodic table; react with water to form alkali (basic) solutions.

Alkali Earth Metals - belong to Group II of the periodic table; not as reactive as the alkali metals.

Transition Metals - found in the middle of the periodic table; most reactive metals are at the top.

Halogens - belong to Group VII of the periodic table; form salts when reacted with metals.

Noble (Inert) Gases - belong to Group VIII of the periodic table; do not react with other elements.

Allotrope - a different form of the same element.

Ion - an atom (or group of atoms) that has a charge due to the gain or loss of electrons.

Valency - the charge on an ion.

Ionic Bond - the electrostatic attraction between oppositely charged ions.

Covalent Bond - formed by the sharing of two electrons between two atoms.

Covalent Network - atoms bond to each other in a 3-dimensional arrangement; elements in Group IV do this.

Solute - substance that dissolves.

Solvent - the substance that does the dissolving.

Solution - formed when one substance dissolves into another.

Precipitate - a solid that forms when two ionic solutions are mixed, forming an insoluble substance.

Spectator Ions - not directly involved in forming a precipitate.

Ionic Equation - shows only the ions involved in a reaction.